

Unit Four: Genetics

4.1. The genetic materials 4.2. The structure and function of DNA and RNA

After the successful completion of this section, the student will be able to:

- Define genetics, genes, chromosomes, DNA, and RNA.
- Describe DNA and RNA

4.2. The structure and function of DNA and RNA

After the successful completion of this section, the student will be able to:

- Describe the structures of DNA and RNA.
- Draw and label the structures of DNA and RNA.
- Describe the process of DNA replication.
- Explain the functions of DNA and RNA.
- Compare DNA with RNA with respect to structure and function.

4.3 The process of cell division

After the successful completion of this section, the student will be able to:

- Describe cell cycle
- Define mitosis and meiosis
- Describe the three stages of interphase
- Mention stages in mitosis and meiosis
- Explain each stages of mitosis and meiosis
- Compare mitosis and meiosis

4.4 Protein synthesis

After the successful completion of this section, the student will be able to:

- Describe the process of protein synthesis
- Differentiate between transcription and translation
- Explain the role of DNA and RNAs in protein synthesis
- Read amino acids from genetic code

4.5 Mendelian inheritance

After the successful completion of this section, the student will be able to:

- Define inheritance.
- Describe Mendelian cross.
- Explain monohybrid cross.
- Differentiate between phenotype and genotype.
- Differentiate between homozygous and heterozygous genotypes.
- Explain the test cross.
- Describe Mendelian laws of dominance and laws of segregation

4.6 Sex determination

At the end of this section, the student will be able to:

- Identify sex chromosomes
- Explain how sex is determining

4.7 Non-Mendelian inheritance

After the successful completion of this section, the student will be able to:

- Define non-Mendelian inheritance.
- Differentiate between dominance, incomplete dominance and codominance.
- Describe ABO blood groups.
- Explain inheritance of blood group.
- Describe blood transfusion.
- Explain Rh blood group and inheritance
- Describe sex-linked inheritance.
- Mention environmental factors that affect the phenotype of organisms

4.8 Human pedigree analysis and its importance

After the successful completion of this section, the student will be able to:

- Define pedigree
- Describe a pedigree using heritable traits.
- Explain the importance of pedigree analysis.

4.9 Genetic disorders

After the successful completion of this section, the student will be able to:

- List common genetic disorders in humans
- Differentiate between single-gene, chromosomal and multifactorial disorders
- Explain the difference between autosomal dominant, recessive and x-linked disorders
- Give examples of genetic disorders in humans.

4.10 Genetic testing and counseling

After the successful completion of this section, the student will be able to:

- Define genetic testing
- Explain genetic counseling
- List the importance of genetic testing and counseling

4.11 Gene therapy

After the successful completion of this section, students will be able to:

- Define gene therapy
- Mention the types of gene therapy
- Explain the types of gene therapy

4.12 Breeding

After the successful completion of this section, the student will be able to:

- Define breeding
- Explain breeding in animals and plants
- Mention breeding mechanisms
- Identify indigenous knowledge used in breeding
- Explain indigenous selection and hybridization mechanisms used by local people

4.13 Bioinformatics introduction

After the successful completion of this section, the student will be able to:

- Define bioinformatics
- Describe the role of bioinformatics in biology